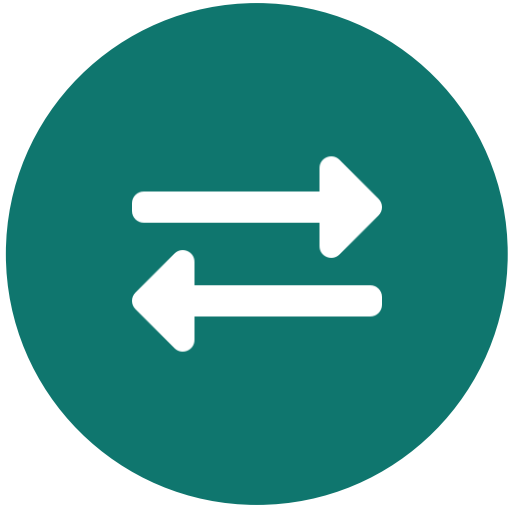
OCTAL CONVERSIONS

Octal → Decimal

Octal → Binary

Octal → Hexadecimal



Octal Number System

Definition:

Octal number system uses 8 digits. One octal digit represents exactly 3 binary bits.

It is convenient for representing binary numbers in shorter form.

Digits:

0	1	2	3	4	5	6	7

Base (Radix): 8

Representation:

$(567)_8$ $(144)_8$ $(31.6)_8$ $(113.2)_8$

Octal → Decimal | $(65)_8$

Without point

Position :	1	0
Octal :	6	5

Value:

$6\times(8)^1 + 5\times(8)^0$

$$48 + 5 = (53)_{10}$$

Octal → Decimal: Method

1. Write the place value (power of 8) under every octal digit.
2. Multiply each digit by its place value.
3. Add all the results to get the decimal number.

$$\text{Decimal} = \Sigma \left(\text{digit} \times 8^{\text{position}} \right)$$

Octal → Decimal | $(12.4)_8$

With point

Position :	1	0	.	–1
Octal :	1	2	.	4

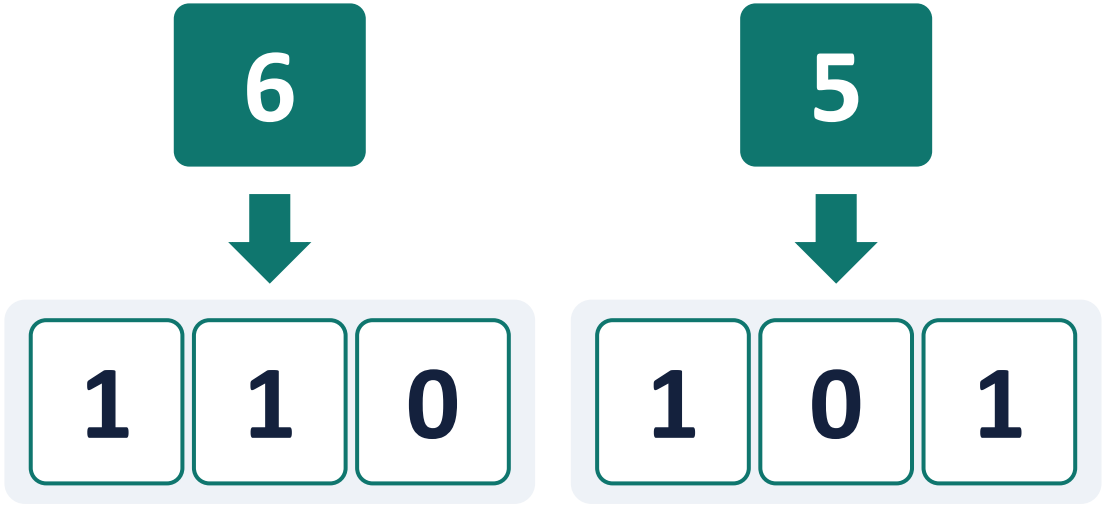
Value:

$$1\times(8)^1 + 2\times(8)^0 + 4\times(8)^{-1}$$

$$8 + 2 + 0.5 = (10.5)_{10}$$

Octal → Binary | $(65)_8$

Without point



3-bit table (the values used in this example are highlighted)

0	1	2	3	4	5	6	7
000	001	010	011	100	101	110	111

Each octal digit becomes 3 bits.

Answer: $(110101)_2$

Octal → Binary: Method

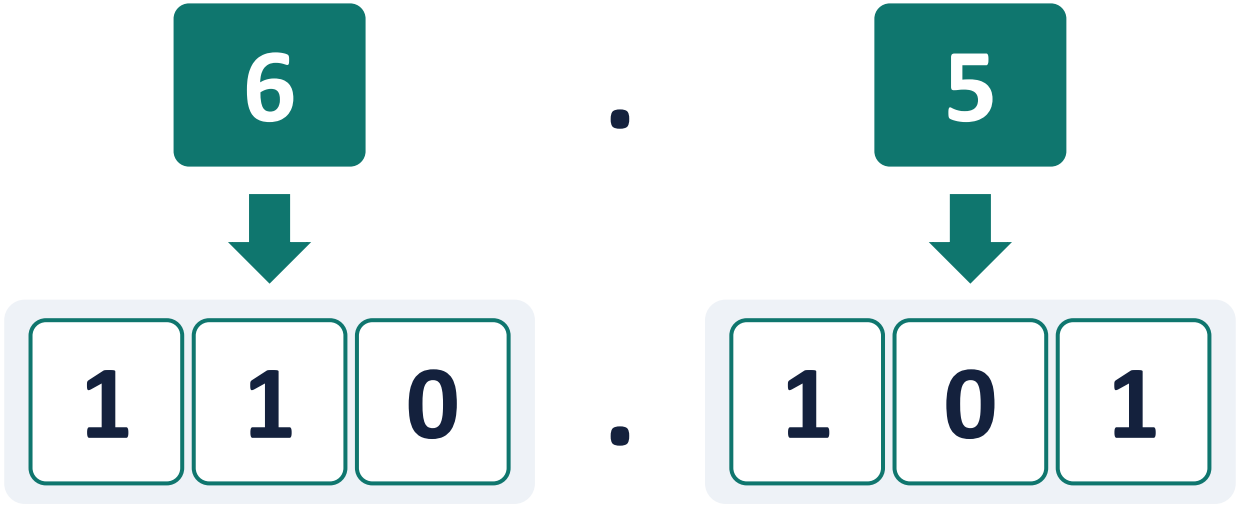
1. Take every octal digit separately.
2. Write the 3-bit binary code of each digit (3-bit table made earlier).
3. Join all the groups and drop the leading zeros.

Lookup table: octal digit → 3 bits

0	1	2	3	4	5	6	7
000	001	010	011	100	101	110	111

Octal → Binary | $(6.5)_8$

With point



3-bit table (the values used in this example are highlighted)

0	1	2	3	4	5	6	7
000	001	010	011	100	101	110	111

Each octal digit becomes 3 bits.

Answer: $(110.101)_2$

Octal → Hexadecimal | $(375)_8$

Without point · Step 1 of 2

Octal

3	7	5

Binary (3 bits each)

0	1	1	1	1	1	1	0	1

Join all the bits

0	1	1	1	1	1	1	0	1

3-bit table (the values used in this example are highlighted)

0	1	2	3	4	5	6	7
000	001	010	011	100	101	110	111

Grey zeros are dropped in the next step.

Octal → Hexadecimal | $(375)_8$

Without point · Step 2 of 2

Combined binary

1 1 1 1 1 1 0 1

Regroup (4 bits each)

1 1 1 1 | 1 1 0 1

Hexadecimal

F D

Answer: $(FD)_{16}$

4-bit table (the values used in this example are highlighted)

0000 **0**	0001 **1**	0010 **2**	0011 **3**	0100 **4**	0101 **5**	0110 **6**	0111 **7**
1000 **8**	1001 **9**	1010 **A** = 10	1011 **B** = 11	1100 **C** = 12	1101 **D** = 13	1110 **E** = 14	1111 **F** = 15

Octal → Hexadecimal: Method

1. Convert every octal digit into 3 bits.
2. Join all the bits together, then make groups of 4 bits from the binary point.
3. Replace each group of 4 bits with its hex digit (A = 10 … F = 15).

Lookup table: 4 bits → hex digit (= decimal value)

0000	0001	0010	0011	0100	0101	0110	0111
0	**1**	**2**	**3**	**4**	**5**	**6**	**7**
= 0	= 1	= 2	= 3	= 4	= 5	= 6	= 7
1000	1001	1010	1011	1100	1101	1110	1111
8	**9**	**A**	**B**	**C**	**D**	**E**	**F**
= 8	= 9	= 10	= 11	= 12	= 13	= 14	= 15

Octal → Hexadecimal | $(6.5)_8$

With point · Step 1 of 2

Octal 6 . 5

Binary (3 bits each) 1 1 0 . 1 0 1

Join all the bits 1 1 0 . 1 0 1

3-bit table (the values used in this example are highlighted)

0	1	2	3	4	5	6	7
000	001	010	011	100	**101**	**110**	111

Octal → Hexadecimal | $(6.5)_8$

With point · Step 2 of 2

Combined binary

1 1 0 . 1 0 1

Regroup (4 bits each)

0 1 1 0 . 1 0 1 0

Hexadecimal

6 . A

Answer: $(6.A)_{16}$

4-bit table (the values used in this example are highlighted)

0000 0	0001 1	0010 2	0011 3	0100 4	0101 5	**0110 6**	0111 7
1000 8	1001 9	**1010 A = 10**	1011 B = 11	1100 C = 12	1101 D = 13	1110 E = 14	1111 F = 15